

Shayan Majidy

Curriculum vitae

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Quantum computing theorist studying how physical dynamics, error-correcting codes, and algorithms jointly determine the cost of reliable quantum computation.

Quantum error correction; quantum algorithms and architectures; quantum dynamics and thermodynamics.

Research appointments

2026	Senior Research Scientist, QuEra
2026	Associate, Harvard University Adviser: Mikhail D. Lukin
2024–2026	Postdoctoral Fellow, Harvard University Advisers: Mikhail D. Lukin Michael Gullans Funding: <u>Banting Fellowship</u> , 2024–2026

Education

2019–2024	Ph.D. in Physics, University of Waterloo Advisers: Raymond Laflamme Nicole Yunger Halpern Funding: <u>Vanier Scholarship</u> , 2021–2024
2018–2019	M.Sc. in Physics, University of Waterloo Advisers: Raymond Laflamme
2011–2015	B.Sc. in Theoretical Physics, University of Guelph

Selected awards

Competitive fellowships

2024–2026	<u>Banting Postdoctoral Fellowship.</u> Government of Canada; CAD 140,000 over two years. Field-specific ranking: 3rd of 185 applicants.
2021–2024	<u>Vanier Canada Graduate Scholarship.</u> Government of Canada; CAD 150,000 over three years.

Research honours

2025	<u>W.B. Pearson Medal.</u> University of Waterloo; creative doctoral research; normally one award per Science department annually.
2024	<u>John Brodie Memorial Award</u> (co-recipient). Perimeter Institute; recognizes creativity in doctoral research.
2022	<u>IQC Achievement Award.</u> Institute for Quantum Computing; excellence in graduate research.

Publications and preprints

1. **Entangling logical qubits without physical operations**
Selected for talks at [TQC 2026](#) and [QEC 2026](#).
J. M. Koh, A. Gong, A. C. Diaconu, D. B. Tan, A. A. Geim, M. J. Gullans, N. Y. Yao, M. D. Lukin, and **S. Majidy**
Preprint · [arXiv:2601.20927](#) (2026)
2. **A fault-tolerant neutral-atom architecture for universal quantum computation**
D. Bluvstein, A. A. Geim, S. H. Li, S. J. Evered, J. P. Bonilla Ataides, G. Baranes, A. Gu, T. Manovitz, M. Xu, M. Kalinowski, **S. Majidy**, C. Kokail, N. Maskara, E. C. Trapp, L. M. Stewart, S. Hollerith, H. Zhou, M. J. Gullans, S. F. Yelin, M. Greiner, V. Vuletić, M. Cain, and M. D. Lukin
Nature **649**, 39–46 (2026) · [Journal](#) · [arXiv](#)
3. **Scalable and fault-tolerant preparation of encoded k -uniform states**
S. Majidy, D. Hangleiter, and M. J. Gullans
Physical Review A **112**, 042409 (2025) · [Journal](#) · [arXiv](#)
4. **Thermodynamic limitations on fault-tolerant quantum computing**
M. Bilokur, S. Gopalakrishnan, and **S. Majidy**
Preprint · [arXiv:2411.12805](#) (2024)
5. **Noncommuting charges can remove non-stationary quantum many-body dynamics**
S. Majidy
Nature Communications **15**, 8246 (2024) · [Journal](#) · [arXiv](#)
6. **Addressing misconceptions in university physics: A review and experiences from quantum physics educators**
S. Majidy
Preprint · [arXiv:2405.20923](#) (2024); revised 2025
7. **Noncommuting conserved charges in quantum thermodynamics and beyond**
S. Majidy, W. F. Braasch Jr., A. Lasek, T. Upadhyaya, A. Kalev, and N. Yunger Halpern
Nature Reviews Physics **5**, 689–698 (2023) · [Journal](#) · [arXiv](#)
8. **Critical phase and spin sharpening in SU(2)-symmetric monitored quantum circuits**
S. Majidy, U. Agrawal, S. Gopalakrishnan, A. C. Potter, R. Vasseur, and N. Yunger Halpern
Physical Review B **108**, 054307 (2023) · [Journal](#) · [arXiv](#)
9. **A unification of the coding theory and OAQEC perspective on hybrid codes**
S. Majidy
International Journal of Theoretical Physics **62**, 177 (2023) · [Journal](#) · [arXiv](#)
10. **Non-Abelian symmetry can increase entanglement entropy**
S. Majidy, A. Lasek, D. A. Huse, and N. Yunger Halpern
Physical Review B **107**, 045102 (2023) · [Journal](#) · [arXiv](#)
11. **How to build Hamiltonians that transport noncommuting charges in quantum thermodynamics**
N. Yunger Halpern and **S. Majidy**
npj Quantum Information **8**, 10 (2022) · [Journal](#) · [arXiv](#)
12. **Detecting violations of macrorealism when the original Leggett-Garg inequalities are satisfied**
S. Majidy, J. J. Halliwell, and R. Laflamme
Physical Review A **103**, 062212 (2021) · [Journal](#) · [arXiv](#)
13. **Exploration of an augmented set of Leggett-Garg inequalities using a noninvasive continuous-in-time velocity measurement**
S. Majidy, H. Katiyar, G. Anikeeva, J. Halliwell, and R. Laflamme
Physical Review A **100**, 042325 (2019) · [Journal](#) · [arXiv](#)

Textbook

Building Quantum Computers: A Practical Introduction

S. Majidy, C. Wilson, and R. Laflamme
Cambridge University Press (2024).

Lead author. Textbook on quantum computing hardware for advanced undergraduates and beginning graduate students.

Research mentoring

Graduate research mentoring. Harvard University; University of Maryland; ETH Zürich; University of Waterloo.

Undergraduate research mentoring. Princeton University; University of Waterloo.

Research talks

Invited conference talks (1)

Achieving the limits of automorphism gates

22 Oct 2026 [Princeton Quantum Technology Conference \(PQTC 2026\)](#), Princeton University

Contributed conference talks (2)

Entangling logical qubits without physical operations

2 Sep 2026 [Theory of Quantum Computation, Communication and Cryptography \(TQC 2026\)](#), Sherbrooke

11 Jun 2026 [Quantum Error Correction \(QEC 2026\)](#), Santa Barbara

Colloquia (5)

Fault-tolerant quantum computing: From fundamental limits to scalable architectures

30 Apr 2026 École de technologie supérieure, Montréal

29 Apr 2026 University of Waterloo

16 Apr 2026 University of Alberta

20 Mar 2026 University of British Columbia, Okanagan

15 Jan 2026 Université de Sherbrooke, Institut Quantique

Invited seminars (23)

Entangling logical qubits without physical operations

26 Jun 2026 [Hon Hai \(Foxconn\) Research Institute](#), Taipei

19 May 2026 [Virginia Tech](#), VTQ Seminar

13 Apr 2026 [Inria](#), QASAR Seminar (online)

4 Mar 2026 [University of Maryland](#), QuICS Seminar · [\[Recording\]](#)

7 Jan 2026 Perimeter Institute, Quantum Information Seminar · [\[Recording\]](#)

Architectural mechanisms of a universal fault-tolerant quantum computer

3 Oct 2025 [University of Toronto](#), CQIQC Seminar · [\[Recording\]](#)

A noncommuting charge puzzle

8 Jan 2025 Perimeter Institute, Quantum Information Seminar · [\[Recording\]](#)

28 Jun 2024 Harvard University, Harvard Quantum Initiative Seminar

Noncommuting charges can increase entanglement and induce critical dynamics

- 16 Jan 2024 [Yale University](#), Yale Quantum Institute Seminar
29 Nov 2023 Perimeter Institute, Quantum Information Seminar · [\[Recording\]](#)
22 Sep 2023 Princeton University, Theory Seminar

Monitored quantum circuits with noncommuting conserved quantities

- 15 Sep 2023 IBM, Qiskit Seminar (online) · [\[Recording\]](#)

Non-Abelian symmetry can increase entanglement entropy

- 28 Mar 2023 Université de Sherbrooke, Institut Quantique / InfoQ Seminar
24 Mar 2023 McGill University, Special INTRIQ/CPM Seminar
24 Feb 2023 Stanford University, Institute for Theoretical Physics Seminar
22 Feb 2023 University of California, Berkeley, Pitzer Center Seminar
22 Feb 2023 [University of California, Berkeley](#), Redwood Seminar
7 Feb 2023 Universität Würzburg, Quantum Field Theory and Gravity Seminar (online)
3 Feb 2023 [University of Toronto](#), CQIQC Seminar

Noncommuting charges: Bridging theory to experiment

- 18 Aug 2022 Australian Institute of Physics, Theoretical Physics Seminar (online)
2 Aug 2022 University of Maryland, RQS Seminar

An introduction to quantum thermodynamics

- 1 Dec 2021 Mila Seminar (online)

Noncommuting charges: Bridging theory to experiment

- 9 Jun 2021 University of Bristol, Quantum Information Theory Seminar (online)

Other talks (7)

Entangling logical qubits without physical operations

- 20 Aug 2026 [CQIQC-XI](#), Toronto

Architectural mechanisms of a universal fault-tolerant quantum computer

- 8 Oct 2025 [Year of Quantum Across Canada](#), Waterloo · [\[Recording\]](#)

Non-Abelian symmetry can increase entanglement entropy

- 26 Aug 2024 [CQIQC-X](#), University of Toronto
19 Jul 2023 Raymond Laflamme's 60th Birthday Conference, University of Waterloo
22 Jun 2023 RQS Annual Workshop, University of Maryland
8 Dec 2022 [Quantum Non-Markovianity 2022](#) (online)

Noncommuting charges: Bridging theory to experiment

- 22 Jul 2022 Information Engines at the Frontiers of Nanoscale Thermodynamics, Telluride Science Research Center

Teaching and educational development

Independent course instruction

Fall 2022 **Sessional Lecturer**, University of Waterloo.
Introduction to Implementation of Quantum Information Processing, fourth-year course.
Independently taught the course; redesigned the syllabus, teaching methods, and assessments; delivered lectures and coordinated teaching assistants.
Course evaluation: 4.8/5 · 97% response rate.

Teaching assistantships

2020–2022 **Teaching Assistant**, University of Waterloo.
Quantum Information Processing Devices (Winter 2020, 2021, 2022); *Introduction to Implementation of Quantum Information Processing* (Fall 2021); *Electricity and Magnetism 1* (Winter 2021).
Led discussion sessions, graded assignments, and held office hours.

Educational development

Jan–May 2022 **Graduate Educational Developer**, Centre for Teaching Excellence.
Jan–Dec 2021 **TA Workshop Facilitator**, Centre for Teaching Excellence.
Across both roles, designed and facilitated workshops, conducted teaching observations and microteaching sessions, and provided individual feedback to graduate instructors.
Qualification: [Certificate in University Teaching](#), University of Waterloo.

Professional service and outreach

Peer review

Journal reviewing (43 reviews). *PRX Quantum* (8); *Physical Review Letters* (4); *Nature Communications* (1); *npj Quantum Information* (3); *Physical Review Research* (2); *Physical Review A* (24); *Physical Review E* (1).
Conference subreviewing. QIP 2026; QEC 2025; QCTIP 2025; TQC 2024.

Scientific event organization

2023 **Organizer**, Raymond Laflamme’s 60th Birthday Conference.
University of Waterloo.
2021 **Seminar organizer**, Quantum Steampunk Seminars. University of Maryland.
2020 **Organizing committee**, Canadian Graduate Quantum Conference. University of Waterloo.

Selected outreach and public engagement

Founder, Unentangled. Built a nonprofit bringing current research into high-school classrooms. At its peak, the program served more than 1,000 students annually with a team of 10 volunteers. [\[Documentary\]](#)
Public engagement: Invited panelist, *Quantum + The Near Future*, Quantum Connections (2024); interview, CBC Radio’s *Quirks & Quarks* (30 Dec 2023).
Research coverage: Phantom codes featured in [New Scientist](#).